Code-Layout, Readability and Reusability

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Project Name	CaptionAI: AI-Powered Social Media Caption Generator

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I organized my codebase to ensure it was clean, readable, maintainable, and reusable. This made the project easier to understand, modify, maintain, and collaborate on.

S.No	Quality Principle	Implementation Details / Description	Specific Project Examples (Files/Folders)
1	Folder/Directory Structure	The project is organized into separate folders for frontend and backend development. HTML templates, CSS, JavaScript, and Python files are placed in their respective directories to improve project organization and maintenance.	`templates/index.html`, `static/css/style.css`, `static/js/main.js`, `app.py`
2	Naming Conventions	Meaningful and consistent names are used for variables, functions, and files. Python code follows the `snake_case` convention, while JavaScript uses `camelCase` for better readability.	`copyToClipboard()`, `submitBtn`, `generateCaption()`, `app.py`
3	Code Readability & Comments	The source code is properly formatted with clear indentation and comments wherever required. This makes the code easier to understand, debug, and maintain.	Comments in `main.js` for loader, API request, and copy functionality.

4	Modularity & Reusability	Reusable functions are created to avoid code duplication. Common tasks such as generating captions and copying text are handled through separate functions.	`copyToClipboard()` function and reusable JavaScript functions in `main.js`.
5	Formatting & Project Configuration	Consistent formatting and indentation are maintained throughout the project. The `requirements.txt` file manages project dependencies, while the `.gitignore` file prevents unnecessary or sensitive files from being uploaded to GitHub.	`requirements.txt`, `.gitignore`, Visual Studio Code formatting.
6	Error Handling & Security	Basic error handling is implemented to display appropriate messages when API requests fail. Sensitive information such as the Groq API key is stored securely in the `.env` file instead of hardcoding it in the source code.	`.env`, `fetch().catch()` in `main.js`, exception handling in `app.py`.